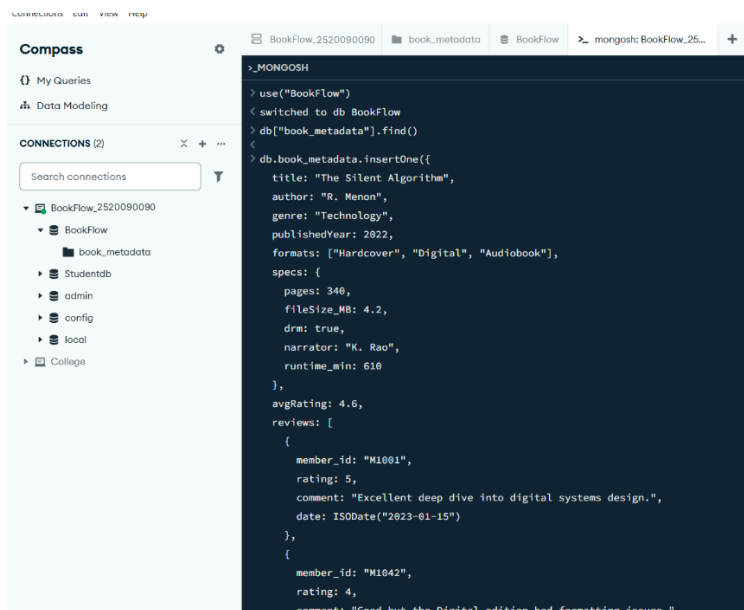


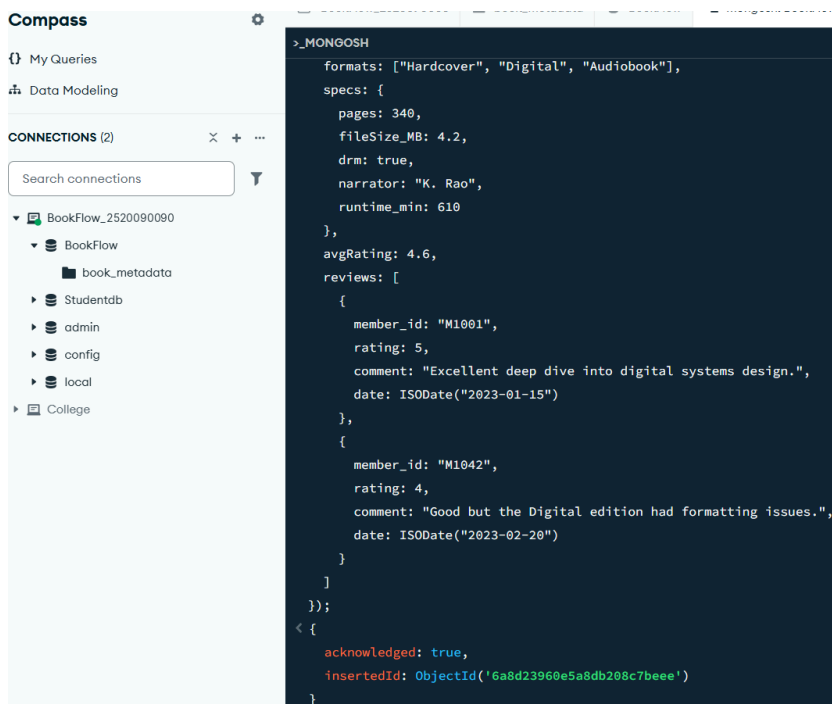
# WEEK-3 Practical

2520090090-Ch.Sai Vihamsh

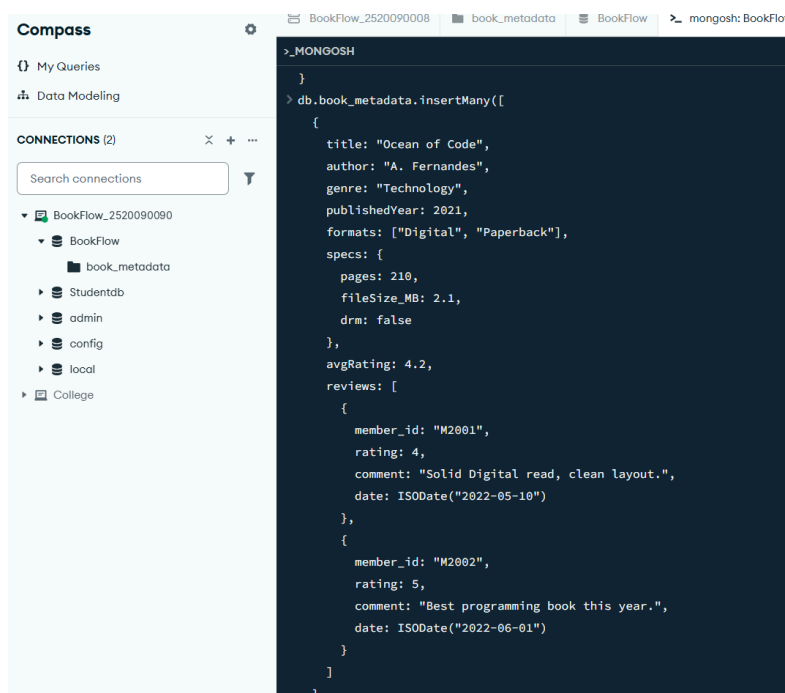
## Task-1:

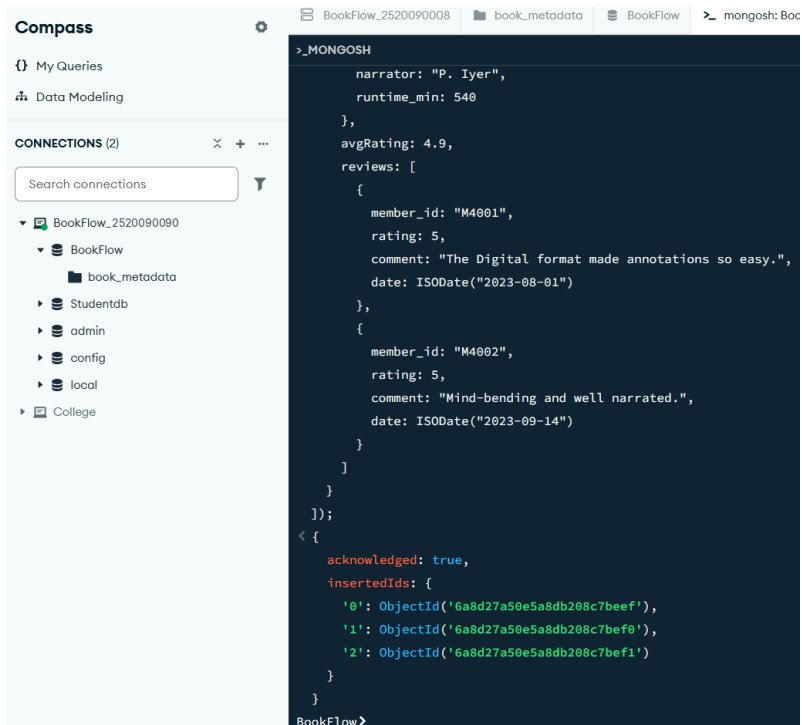
Inserting the First Book Document



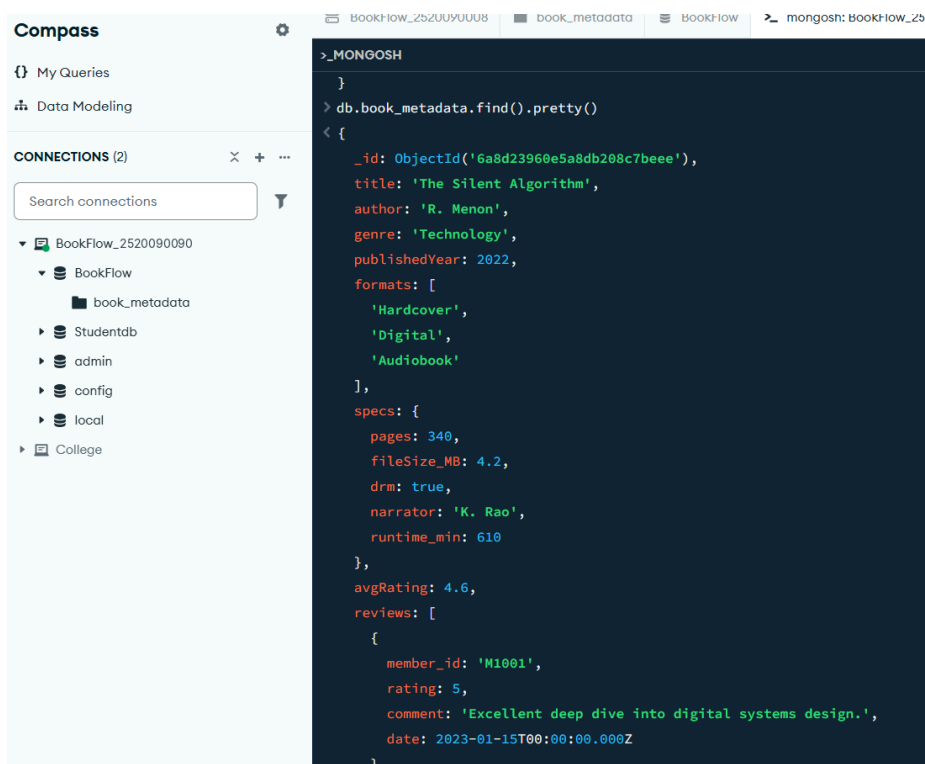


## Inserting Multiple Book Documents



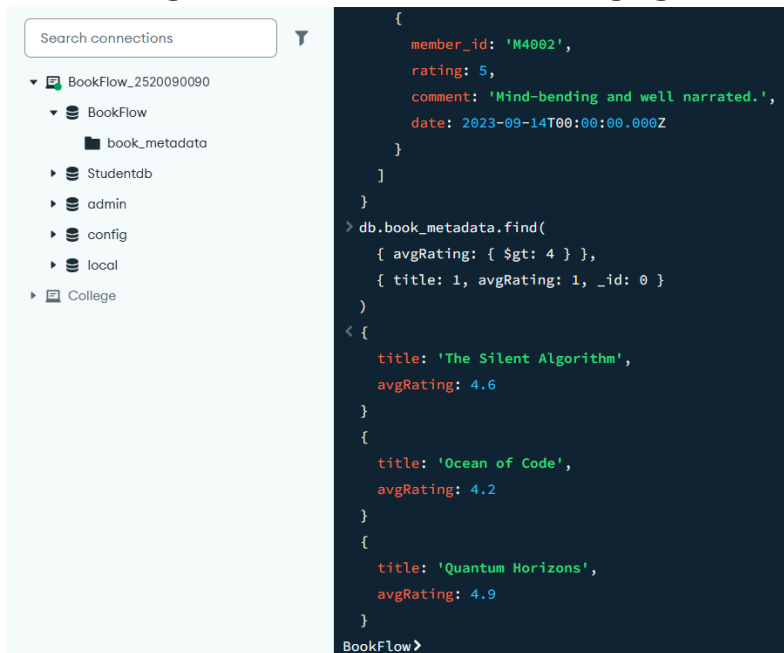


## Displaying the Book Metadata Collection



## TASK – 2

### 2.1 Finding Books Rated Above 4 Using \$gt



The screenshot shows a MongoDB interface with a left sidebar displaying a project structure. The main area shows a query and its results.

**Search connections:**

- BookFlow\_2520090090
  - BookFlow
    - book\_metadata
    - Studentdb
    - admin
    - config
    - local
  - College

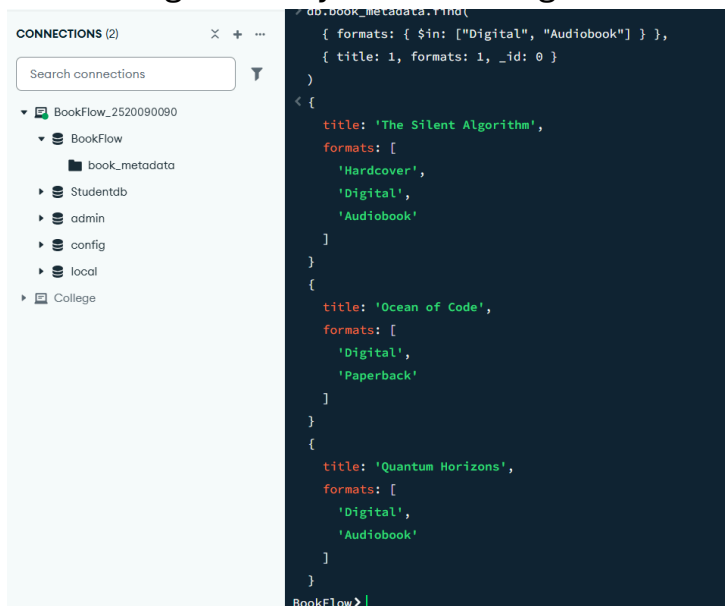
**Query:**

```
> db.book_metadata.find(
  { avgRating: { $gt: 4 } },
  { title: 1, avgRating: 1, _id: 0 }
)
```

**Results:**

```
< {
  title: 'The Silent Algorithm',
  avgRating: 4.6
}
{
  title: 'Ocean of Code',
  avgRating: 4.2
}
{
  title: 'Quantum Horizons',
  avgRating: 4.9
}
BookFlow>
```

### 2.2 Finding Books by Format Using \$in



The screenshot shows a MongoDB interface with a left sidebar displaying a project structure. The main area shows a query and its results.

**Search connections:**

- BookFlow\_2520090090
  - BookFlow
    - book\_metadata
    - Studentdb
    - admin
    - config
    - local
  - College

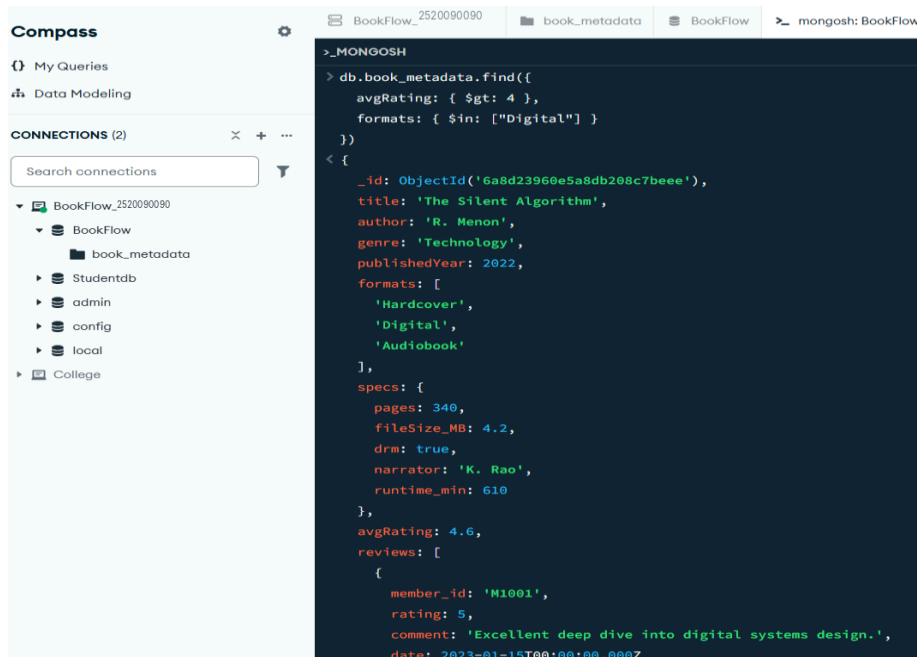
**Query:**

```
> db.book_metadata.find(
  { formats: { $in: ["Digital", "Audiobook"] } },
  { title: 1, formats: 1, _id: 0 }
)
```

**Results:**

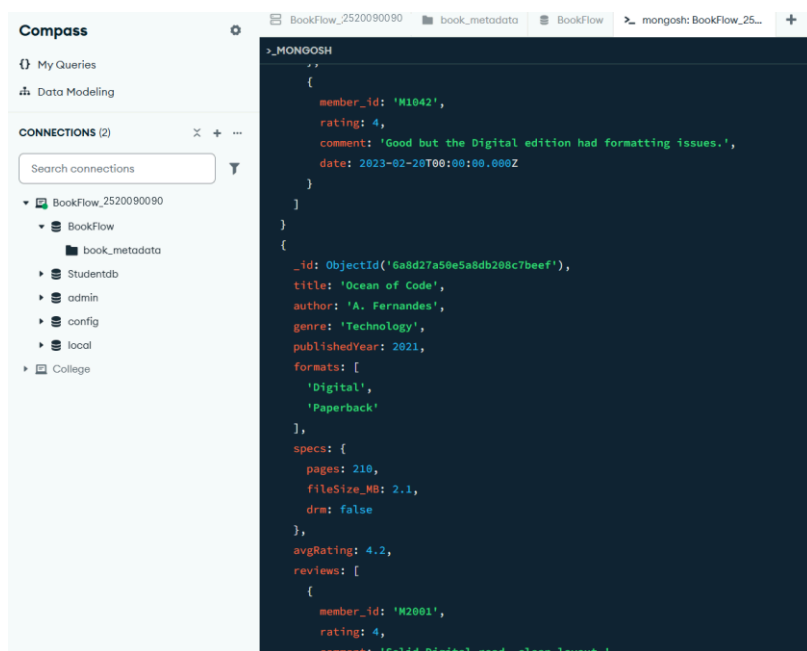
```
< {
  title: 'The Silent Algorithm',
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ]
}
{
  title: 'Ocean of Code',
  formats: [
    'Digital',
    'Paperback'
  ]
}
{
  title: 'Quantum Horizons',
  formats: [
    'Digital',
    'Audiobook'
  ]
}
BookFlow>
```

## 2.3 Combining Rating and Format Conditions



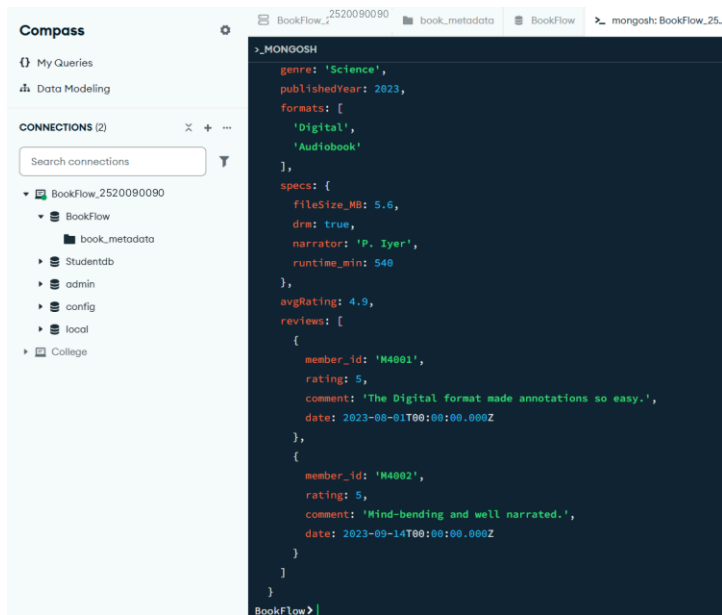
The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (2)' panel lists the database 'BookFlow\_2520090090' and its collections 'BookFlow' and 'book\_metadata'. The 'My Queries' panel is empty. The 'Data Modeling' panel is also empty. The 'mongosh' terminal on the right shows the following query and result:

```
>_MONGOSH
> db.book_metadata.find({
  avgRating: { $gt: 4 },
  formats: { $in: ["Digital"] }
})
< {
  _id: ObjectId('6a8d23960e5a8db208c7beee'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    }
  ]
}
```

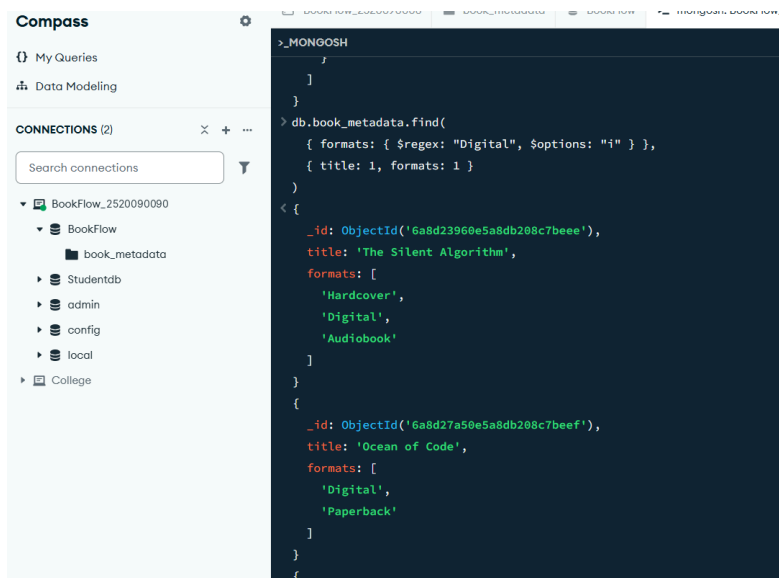


The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (2)' panel lists the database 'BookFlow\_2520090090' and its collections 'BookFlow' and 'book\_metadata'. The 'My Queries' panel is empty. The 'Data Modeling' panel is also empty. The 'mongosh' terminal on the right shows the following query and result:

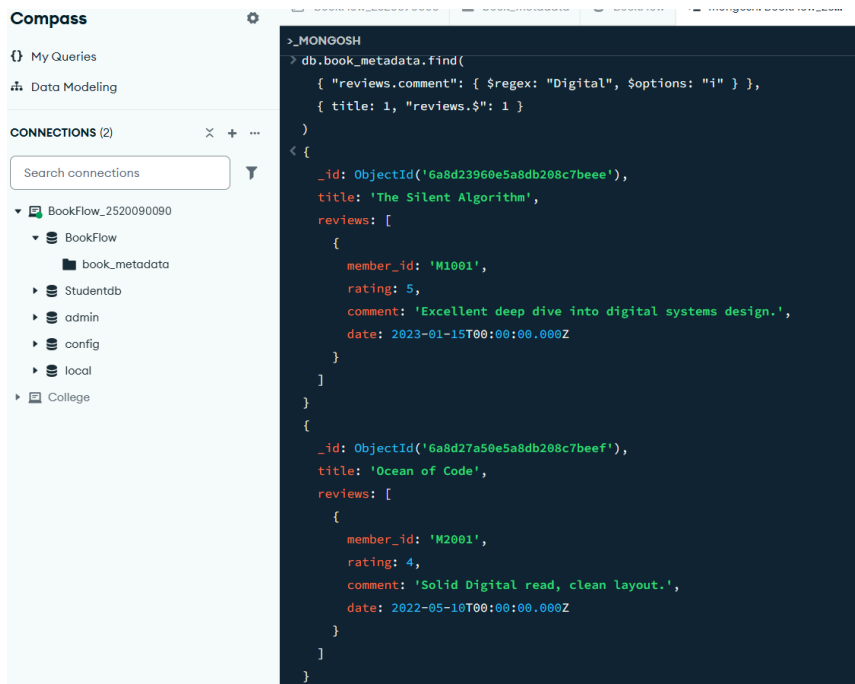
```
>_MONGOSH
> db.book_metadata.find({
  avgRating: { $gt: 4 },
  formats: { $in: ["Digital"] }
})
< {
  _id: ObjectId('6a8d27a50e5a8db208c7beef'),
  title: 'Ocean of Code',
  author: 'A. Fernandes',
  genre: 'Technology',
  publishedYear: 2021,
  formats: [
    'Digital',
    'Paperback'
  ],
  specs: {
    pages: 210,
    fileSize_MB: 2.1,
    drm: false
  },
  avgRating: 4.2,
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read, clean layout.'
    }
  ]
}
```



## Finding Digital Books Using Regular Expression

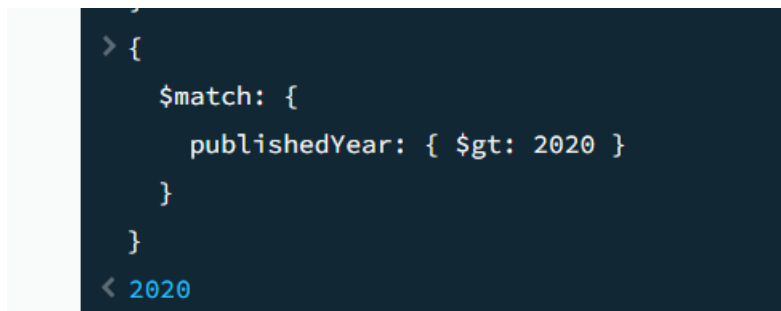


## Searching Reviews Containing "Digital" Using Regex

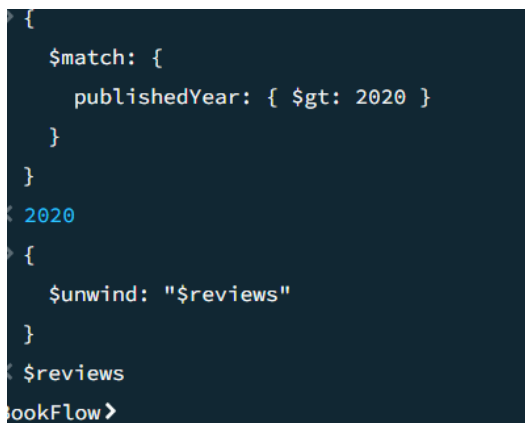


## TASK – 3:

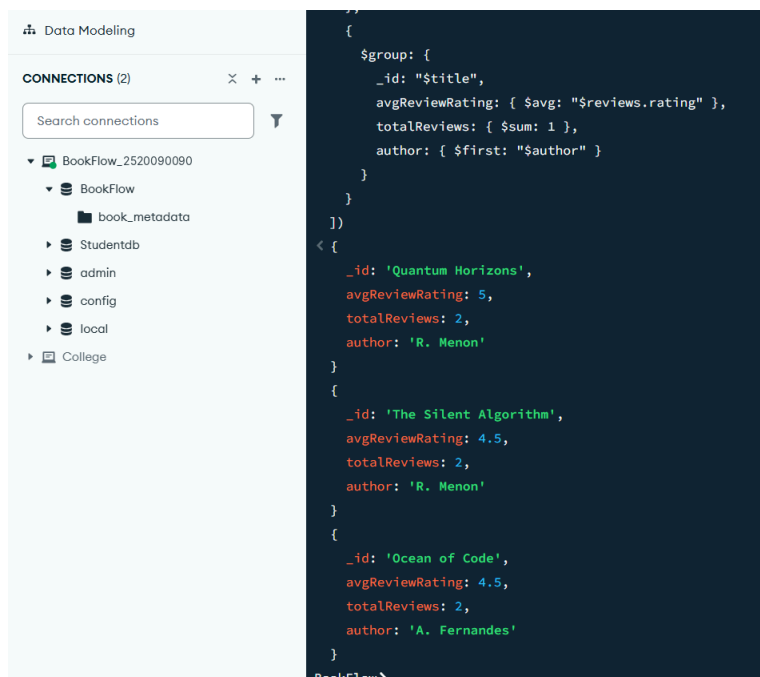
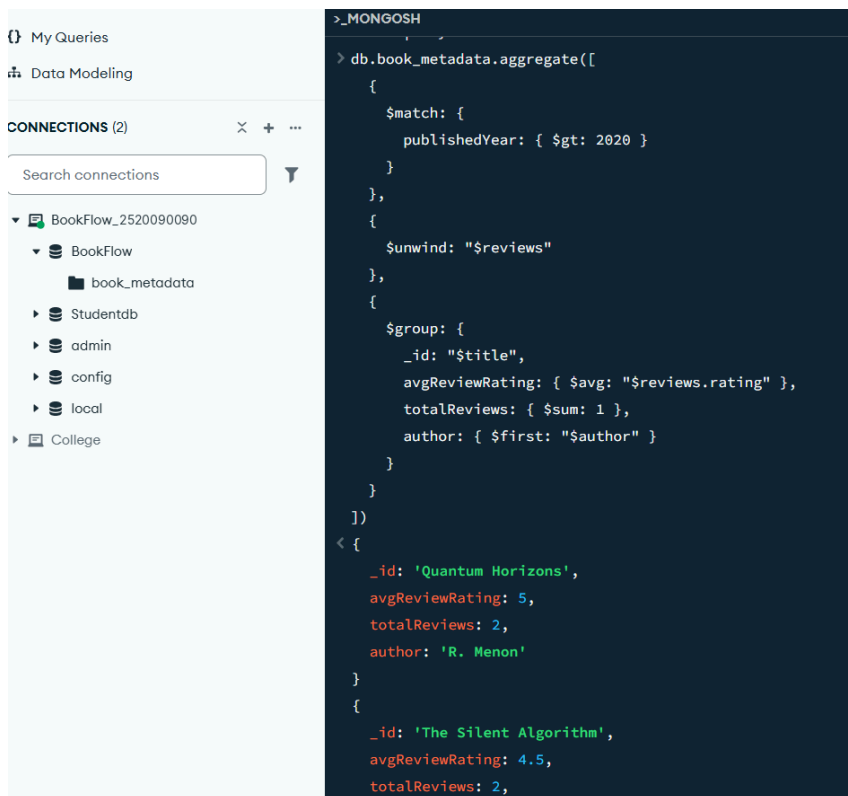
### \$match — Books Published After 2020



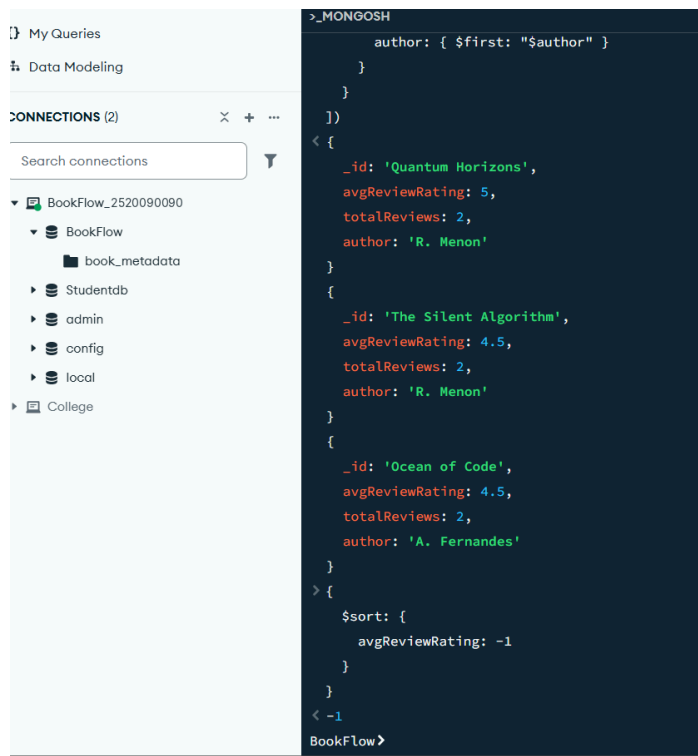
### \$unwind — Expand Reviews



### Grouping Books and Calculating Average Rating Using \$group



## Sorting Books by Average Rating Using \$sort



## Formatting the Top Rated Report Using \$project

Compass

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

BookFlow\_2520090090

- BookFlow
  - book\_metadata
- Studentdb
- admin
- config
- local

College

```
>_MONGOSH
> db.book_metadata.aggregate([
  {
    $match: {
      publishedYear: { $gt: 2020 }
    }
  },
  {
    $unwind: "$reviews"
  },
  {
    $group: {
      _id: "$title",
      avgReviewRating: { $avg: "$reviews.rating" },
      totalReviews: { $sum: 1 },
      author: { $first: "$author" }
    }
  },
  {
    $sort: {
      avgReviewRating: -1
    }
  },
  {
    $project: {
      _id: 0,
      title: "$_id",
      author: 1,
      avgReviewRating: {
        $round: ["$avgReviewRating", 2]
      }
    }
  }
])
```

Compass

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

BookFlow\_2520090090

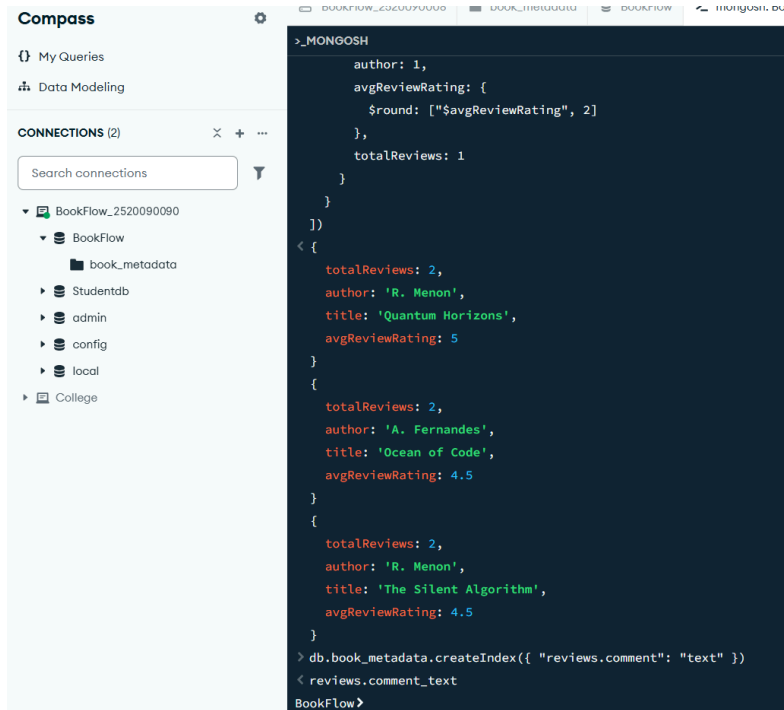
- BookFlow
  - book\_metadata
- Studentdb
- admin
- config
- local

College

```
>_MONGOSH
  _id: 0,
  title: "$_id",
  author: 1,
  avgReviewRating: {
    $round: ["$avgReviewRating", 2]
  },
  totalReviews: 1
}
])
< {
  totalReviews: 2,
  author: 'R. Menon',
  title: 'Quantum Horizons',
  avgReviewRating: 5
}
{
  totalReviews: 2,
  author: 'A. Fernandes',
  title: 'Ocean of Code',
  avgReviewRating: 4.5
}
{
  totalReviews: 2,
  author: 'R. Menon',
  title: 'The Silent Algorithm',
  avgReviewRating: 4.5
}
}
BookFlow>
```

## Task 4: Indexing for Speed

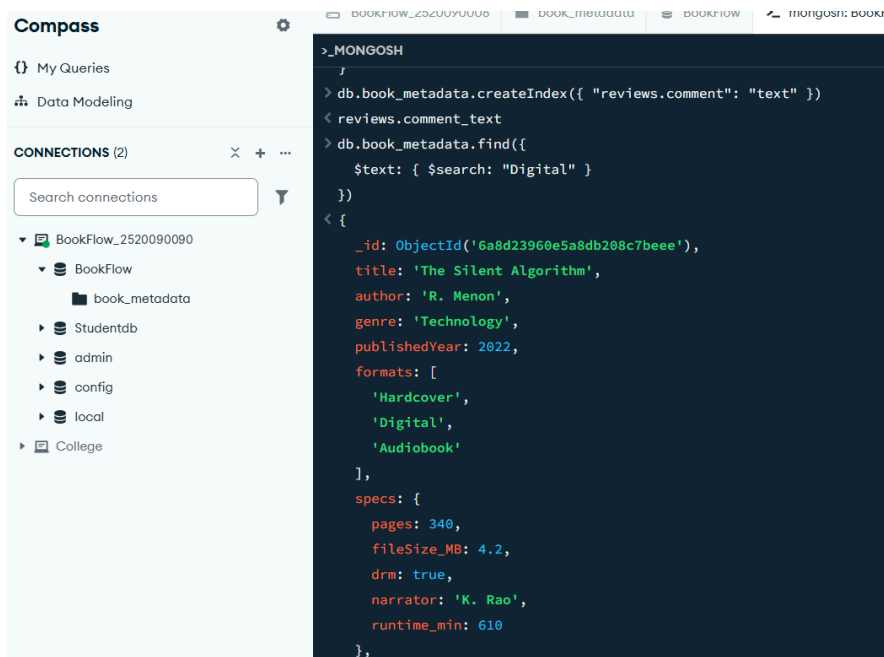
### 4.1 Creating a Text Index on Review Comments



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (2)' panel lists the database 'BookFlow\_2520090090' and its collections, including 'book\_metadata'. The main panel displays a MongoDB shell session with the following commands and results:

```
>_MONGOSH
{
  author: 1,
  avgReviewRating: {
    $round: ["$avgReviewRating", 2]
  },
  totalReviews: 1
}
})
< {
  totalReviews: 2,
  author: 'R. Menon',
  title: 'Quantum Horizons',
  avgReviewRating: 5
}
{
  totalReviews: 2,
  author: 'A. Fernandes',
  title: 'Ocean of Code',
  avgReviewRating: 4.5
}
{
  totalReviews: 2,
  author: 'R. Menon',
  title: 'The Silent Algorithm',
  avgReviewRating: 4.5
}
}
> db.book_metadata.createIndex({ "reviews.comment": "text" })
< reviews.comment_text
BookFlow>
```

### 4.2 Searching Review Comments Using the Text Index



The screenshot shows the MongoDB Compass interface. The main panel displays a MongoDB shell session with the following commands and results:

```
>_MONGOSH
{
}
> db.book_metadata.createIndex({ "reviews.comment": "text" })
< reviews.comment_text
> db.book_metadata.find({
  $text: { $search: "Digital" }
})
< {
  _id: ObjectId('6a8d23960e5a8db208c7beee'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  }
},
```

### 4.3 Verifying Text Index Usage Using .explain()

Compass

{ } My Queries

Data Modeling

CONNECTIONS (2) ✕ + ...

Search connections 🔍

- BookFlow\_2520090090
  - BookFlow
    - book\_metadata
  - Studentdb
  - admin
  - config
  - local
- College

```
>_MONGOSH
}
> db.book_metadata.find(
  $text: { $search: "Digital" }
).explain("executionStats")
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: 'BookFlow.book_metadata',
    parsedQuery: {
      '$text': {
        '$search': 'Digital',
        '$language': 'english',
        '$caseSensitive': false,
        '$diacriticSensitive': false
      }
    },
    indexFilterSet: false,
    queryHash: '4FB85230',
    planCacheShapeHash: '4FB85230',
    planCacheKey: 'F34F3876',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'TEXT_MATCH',
      nss: 'BookFlow.book_metadata',
```

Compass

{ } My Queries

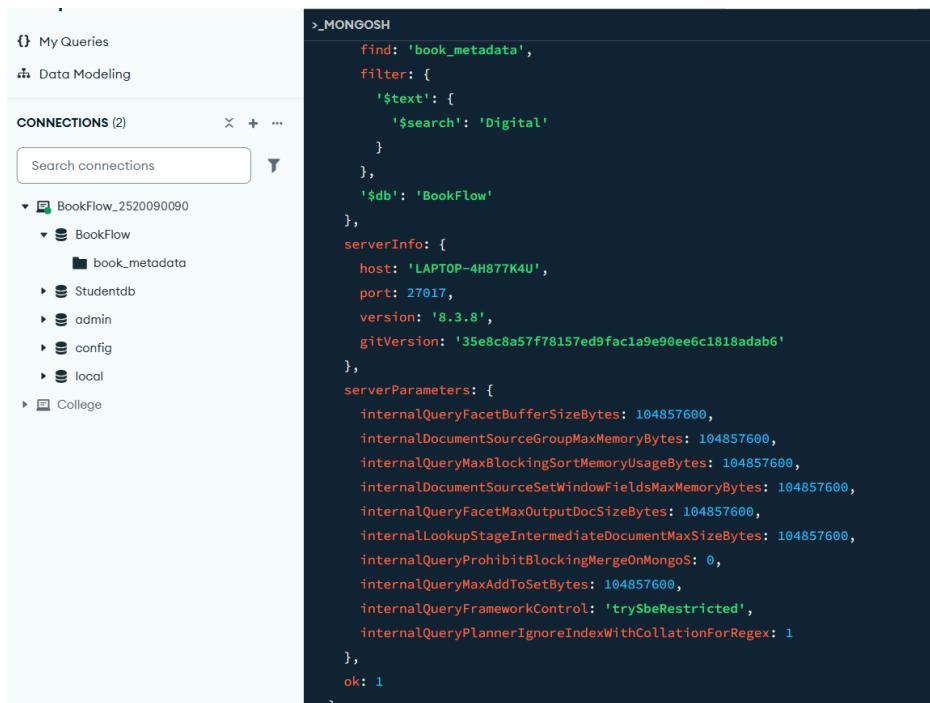
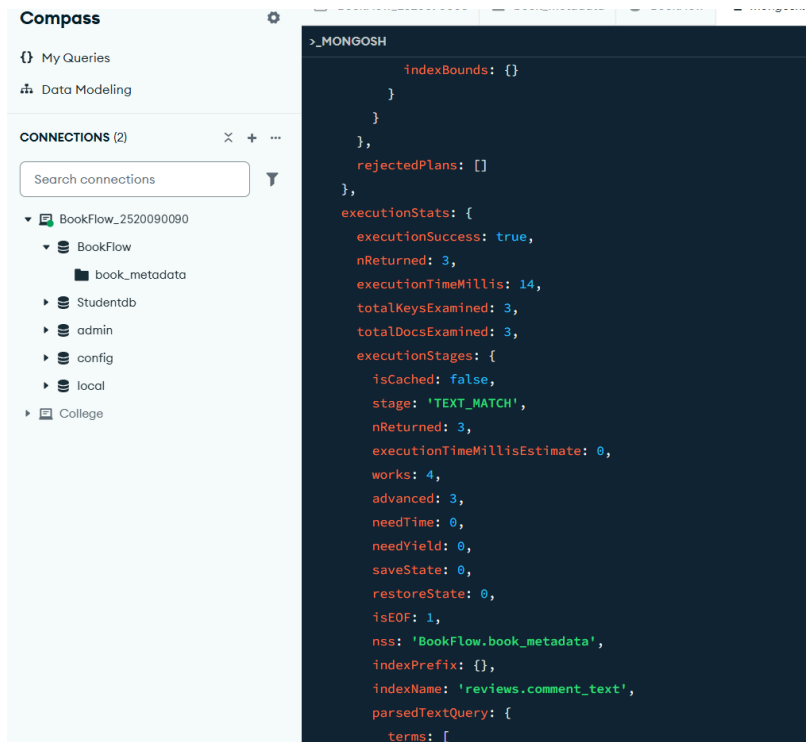
Data Modeling

CONNECTIONS (2) ✕ + ...

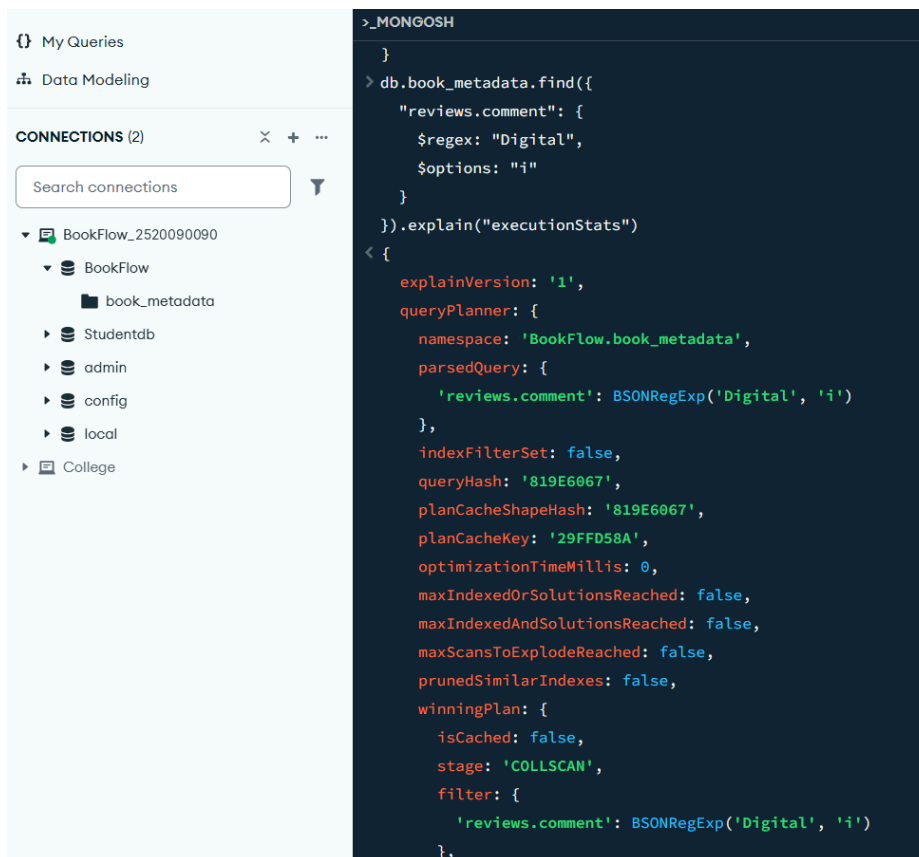
Search connections 🔍

- BookFlow\_2520090090
  - BookFlow
    - book\_metadata
  - Studentdb
  - admin
  - config
  - local
- College

```
>_MONGOSH
nss: 'BookFlow.book_metadata',
indexPrefix: {},
indexName: 'reviews.comment_text',
parsedTextQuery: {
  terms: [
    'digit'
  ],
  negatedTerms: [],
  phrases: [],
  negatedPhrases: []
},
textIndexVersion: 3,
inputStage: {
  stage: 'FETCH',
  nss: 'BookFlow.book_metadata',
  inputStage: {
    stage: 'IXSCAN',
    nss: 'BookFlow.book_metadata',
    keyPattern: {
      _fts: 'text',
      _ftsx: 1
    },
  },
  indexName: 'reviews.comment_text',
  isMultiKey: true,
  isUnique: false,
  isSparse: false,
  isPartial: false,
  indexVersion: 2,
  direction: 'backward',
```



## 4.4 Comparing Regex Search Performance Using .explain()



## 4.5 Creating a Multikey Index on Book Formats

```
> db.book_metadata.createIndex({ formats: 1 })
< formats_1
BookFlow>
```

## 4.6 Creating an Index on Average Rating

```
OK! 1
}
> db.book_metadata.createIndex({ formats: 1 })
< formats_1
> db.book_metadata.createIndex({ avgRating: -1 })
< avgRating_-1
BookFlow> |
```

## 4.7 Verifying Average Rating Index Using .explain()

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

- BookFlow\_2520090090
  - BookFlow
    - book\_metadata
  - Studentdb
  - admin
  - config
  - local
  - College

```
>_MONGOSH
> db.book_metadata.find({
  avgRating: { $gt: 4 }
}).explain("executionStats")
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: 'BookFlow.book_metadata',
    parsedQuery: {
      avgRating: {
        '$gt': 4
      }
    },
    indexFilterSet: false,
    queryHash: 'E21FCB34',
    planCacheShapeHash: 'E21FCB34',
    planCacheKey: '6A4442CC',
    optimizationTimeMillis: 5,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'FETCH',
      nss: 'BookFlow.book_metadata',
      inputStage: {
        stage: 'IXSCAN',
        nss: 'BookFlow.book_metadata',
```

Compass

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

- BookFlow\_2520090090
  - BookFlow
    - book\_metadata
  - Studentdb
  - admin
  - config
  - local
  - College

```
>_MONGOSH
    keyPattern: {
      avgRating: -1
    },
    indexName: 'avgRating_-1',
    isMultiKey: false,
    multiKeyPaths: {
      avgRating: []
    },
    isUnique: false,
    isSparse: false,
    isPartial: false,
    indexVersion: 2,
    direction: 'forward',
    indexBounds: {
      avgRating: [
        '[inf, 4)'
      ]
    }
  },
  rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: 3,
  executionTimeMillis: 14,
  totalKeysExamined: 3,
  totalDocsExamined: 3,
  executionStages: {
```

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

BookFlow\_2520090090

BookFlow

book\_metadata

Studentdb

admin

config

local

College

2\_MONGODB

```
find: 'book_metadata',
filter: {
  avgRating: {
    '$gt': 4
  }
},
'sdb': 'BookFlow'
},
serverInfo: {
  host: 'LAPTOP-4H877K4U',
  port: 27017,
  version: '8.3.8',
  gitVersion: '35e8c8a57f78157ed9fac1a9e90ee6c1818adab6'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}
```